

Практикум по научному письму

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Лабораторная работа 3

Изучение правил написания формул

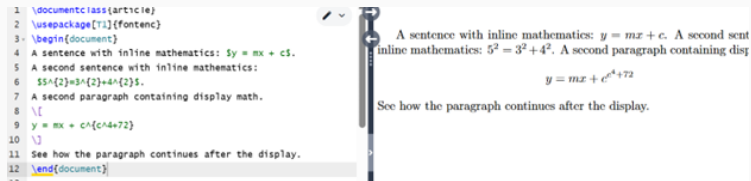


Рис. 1: LaTeX

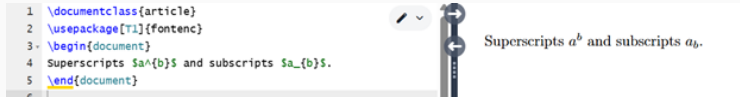


Рис. 2: LaTeX



Рис. 3: LaTeX



Рис. 4: LaTeX

```
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4 \begin{document}
5 Solve the following recurrence for  $n, k \geq 0$ :
6 \begin{align*}
7 Q_{n,0} &= 1 \quad Q_{0,k} = [k=0]; \quad \backslash\backslash
8 Q_{n,k} &= Q_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k},
9 \quad \backslash\quad \text{for } n, k > 0.
10 \end{align*}
11 \end{document}
```

Solve the following recurrence for $n, k \geq 0$:

$$Q_{n,0} = 1 \quad Q_{0,k} = [k=0];$$

$$Q_{n,k} = Q_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k}, \quad \text{for } n, k > 0.$$

Рис. 5: LaTeX

```
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4 \begin{document}
5 AMS matrices.
6 \[
7 \begin{matrix}
8 a & b & c \\
9 d & e & f
10 \end{matrix}
11 \quad
12 \begin{pmatrix}
13 a & b & c \\
14 d & e & f
15 \end{pmatrix}
16 \quad
17 \begin{bmatrix}
18 a & b & c \\
19 d & e & f
20 \end{bmatrix}
21 \]
22 \end{document}
```

AMS matrices.

$$\begin{matrix} a & b & c \\ d & e & f \end{matrix} \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

Рис. 6: LaTeX


```

1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \begin{document}
4 The matrix  $\mathit{ASDDFGGHJKL}$ $.
5
6 The matrix  $\mathbf{ASDDFGGHJKL}$ $.
7
8 The matrix  $\mathsf{ASDDFGGHJKL}$ $.
9
10 The matrix  $\mathtt{ASDDFGGHJKL}$ $.
11
12 The matrix  $\mathbb{ASDDFGGHJKL}$ $.
13
14 The matrix  $\mathrm{ASDDFGGHJKL}$ $.
15
16 The matrix  $\mathtt{\mathsf{ASDDFGGHJKL}}$ $.
17 \end{document}

```

The matrix ASDDFGGHJKL : ".
The matrix ASDDFGGHJ K L : ".
The matrix ASDDFGGHJ K L : ".
The matrix ASDDFGGH J K L : ".
The matrix ASDDFG G H J K L : ".
The matrix ASDDFGGH J K L : ".
The matrix ASDDFGGH J K L : ".

Рис. 7: LaTeX

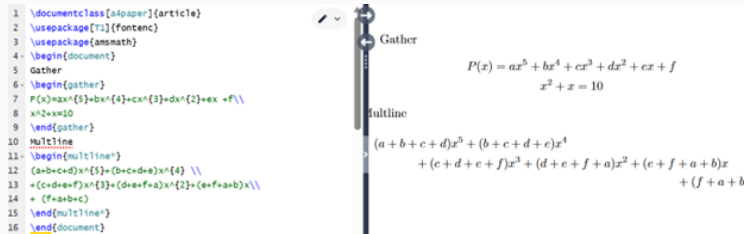
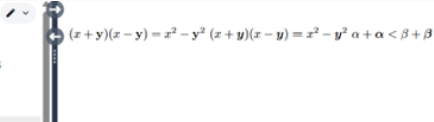


Рис. 8: LaTeX

```
1 \documentclass[a4paper]{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{bm}
4 \begin{document}
5 $(x+\mathbf{y})(x-\mathbf{y})=x^2-y^2-(\mathbf{y})^2$
6 $(x+\bm{y})(x-\bm{y}) \quad \bm{=} \quad x^2-\{\bm{y}\}^2$
7 $\alpha + \bm{\alpha} < \beta + \bm{\beta}$
8 \end{document}
```

The screenshot shows a LaTeX editor interface. On the left, a code editor displays LaTeX source code with line numbers 1 through 8. The code includes document class, font encoding, and the boldsymbol package, followed by several mathematical expressions using boldface commands. On the right, a preview window shows the rendered output of the code, displaying the same mathematical expressions with boldface symbols correctly rendered. A vertical toolbar with icons for undo, redo, and zoom is visible between the code and preview areas.

$(x+y)(x-y) = x^2 - y^2$ $(x+\bm{y})(x-\bm{y}) = x^2 - \bm{y}^2$ $\alpha + \bm{\alpha} < \beta + \bm{\beta}$

Рис. 9: LaTeX

- Изучил новый режим
- Изучил все необходимые формулы и расширения, предоставляемые пакетом `amsmath`

Спасибо за внимание!